Physics 362 - Homework 7

Problem 1: The goal of this problem is to get you familiar with the Zeldovich approximation. We'll do things in the Newtonian limit and assume that there is only matter in the universe ($\Omega_{m,0} = 1$).

Consider a test particle embedded in a dark matter distribution with mass density ρ. This particle has a physical location $\boldsymbol{r}$ and velocity $\boldsymbol{u}$. The equations governing the motion of this particle are simply

$$\frac{d\boldsymbol{r}}{dt} = \boldsymbol{u}, \quad \frac{d\boldsymbol{u}}{dt} = -\frac{\partial \Phi}{\partial \boldsymbol{r}}, \quad \frac{\partial}{\partial \boldsymbol{r}} \cdot \left(\frac{\partial \Phi}{\partial \boldsymbol{r}} \right) = 4\pi G \rho.$$

(a) [3pt] Let $\boldsymbol{x} = \boldsymbol{r}/a$ be our usual comoving coordinates. The physical velocity $\boldsymbol{u}$ then has two components when written in terms of $\boldsymbol{x}$: $\boldsymbol{u} = \dot{\boldsymbol{r}} = a\dot{\boldsymbol{x}} + H\boldsymbol{r}$. The second term is due only to the expansion of the universe while the first term we define as the **peculiar velocity** $\boldsymbol{v} = a\dot{\boldsymbol{x}}$. Similarly we can define a **peculiar potential** $\phi = \Phi + \frac{1}{2}a\ddot{a}\boldsymbol{x}^2$. Show in terms of these variables that our evolution equations becomes

$$\frac{d\boldsymbol{x}}{dt} = \frac{\boldsymbol{p}}{a^2}, \quad \frac{d\boldsymbol{p}}{dt} = -\nabla\phi, \quad \nabla^2\phi = 4\pi G \bar{\rho}_{m,0}\delta/a, \tag{1}$$

where we defined the density contrast δ as $\rho = \bar{\rho}(1+\delta)$, comoving momentum $\boldsymbol{p} = a\boldsymbol{v}$, and let $\nabla = \partial/\partial\boldsymbol{x}$.

Hint: You may need to use the Raychaudhuri equation applied to a matter dominated universe $\ddot{a}/a = -4\pi G\bar{\rho}/3$ which comes from the spatial part of the Einstein equation for a FLRW metric. See e.g. Eq.(2.137) of Baumann's book.

(b) [2pt] The Zeldovich approximation essentially boils down to the ansatz that particles move in straight lines in the direction of their initial displacement:

$$\boldsymbol{x}(t) = \boldsymbol{q} - c(t)\boldsymbol{f}(\boldsymbol{q}). \tag{2}$$

Here $\boldsymbol{q}$ is the initial (Lagrangian) position of the particle and $\boldsymbol{f}(\boldsymbol{q})$ is the direction of the initial displacement determined by the initial conditions at Lagrangian position $\boldsymbol{q}$. We will determine $c(t)$ shortly.

From conservation of mass $\bar{\rho}(1 + \delta(\boldsymbol{x}, t))d^3\boldsymbol{x} = \bar{\rho}d^3\boldsymbol{q}$, show that the density contrast evolves like

$$1 + \delta = \frac{1}{(1 - c(t)\lambda_1)(1 - c(t)\lambda_2)(1 - c(t)\lambda_3)},$$

where λ_i are eigenvalues of the **deformation tensor** $\partial f_i / \partial x_j$.

Commentary: This formula yields a lot of fun insights. For example, if $\lambda_i > 0$ then the DM distribution is collapsing in the direction of the ith eigenvector. You can imagine a blob of matter collapsing along one direction looking like a pancake. This is why some funny people called this "Zeldovich Pancake Theory" back in the day. Once $c(t) = 1/\lambda_i$, the perturbation has collapsed along the corresponding direction and you formed a sheet. If you've collapsed along two direction (e.g. $c(t) > 1/\lambda_1, 1/\lambda_2$) then you've formed a filament, and if you've collapsed along three direction that's a node. From this you can see how the cosmic web emerged pretty naturally in this approximation.

From here on out we will assume that $c\lambda_i \ll 1$ such that we are in regime where linear perturbation theory can be applied

(c) [2pt] First lets figure out the direction these particles will be flying in. Show in linear perturbation theory that $\delta = c(\nabla \cdot \boldsymbol{f})$. Suppose there exists some time t_0 where $c(t_0) = 1$ (you'll see this is possible in the next part) and let $\delta(t_0) = \delta_0$. Show that this implies

$$\boldsymbol{f}(\mathbf{k}) = -i\frac{\mathbf{k}}{k^2}\delta_0(\mathbf{k}).$$

This should make intuitive sense: the direction a particle moves in is set by the initial perturbations at the particle's initial positions.

(d) [3pt] By combining Eq.(1) with (2), show to linear order that $c(t)$ is governed by the differential equation

$$\frac{d^2c}{da^2} + \frac{3}{2a}\frac{dc}{da} = \frac{3}{2a^2}c \Rightarrow c \sim a + (\text{decaying mode}).$$

We can neglect the decaying mode.
Hint: Compute $d^2\boldsymbol{x}/da^2$ to relate $(c, \boldsymbol{f})$ with $\nabla\phi$. Then take the gradient of everything (e.g. $\nabla \cdot d^2\boldsymbol{x}/da^2$). There should be a common $(\nabla \cdot \boldsymbol{f})$ that cancels from both sides.

With this, we have derived the Zeldovich approximation in EdS:

$$\boxed{\boldsymbol{x} = \boldsymbol{q} + a\int \frac{d^3k}{(2\pi)^3}\frac{i\mathbf{k}}{k^2}e^{i\mathbf{k}\cdot\boldsymbol{q}}\delta_0(\mathbf{k}).} \tag{3}$$

(Bonus: Comparing Zeldovich to N-body simulation) [5pt]
I have uploaded to Canvas two files derived from an cosmological N-body simulation with 128^3 particles in a 512 Mpc/h box assuming $\Omega_{m,0} = 1$ (no Λ).

a. `HW07_delta_q_0.npy`: The initial perturbation field δ_0 in $\boldsymbol{q}$ space sam-

pled on a uniform grid spanning the box.

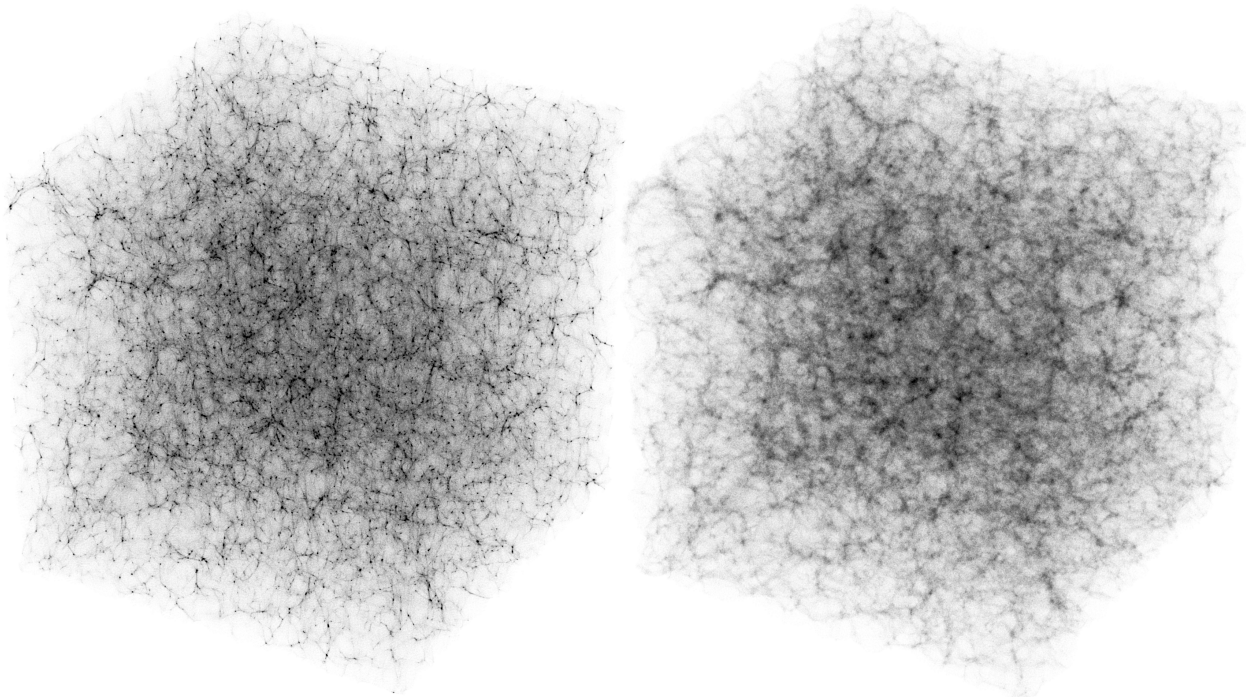
b. **HW07_particle_pos.npy**: Position of all the particles at $z = 0$ after running the full N-body simulation with the initial conditions from (1)[1].

Both of these files can be loaded with **numpy.load**.

```
import numpy as np
delta_q_0 = np.load('HW07_delta_q_0.npy')
pos = np.load('HW07_particle_pos.npy')
```

Use Eq.(3) and **FFT**'s to make a perturbative prediction (Zeldovich) for how the cosmic web should look[2]. Compare this with the non-perturbative prediction from the N-body simulation. For this box, you should get something like this[3]

N-body Zeldovich



[1] If you want positions at any other redshift, let me know. I have them.

[2] Note: you must set $\boldsymbol{f}(\mathbf{k} = 0) = 0$ manually.

[3] There are other fun things to do. For example, you could try smoothing δ_0 and then looking at how the Zeldovich prediction changes. Or you could try making a movie of the cosmic web formation with just Eq.(3).